

Probing magnetic phase transition from neural network wavefunction

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(Dated: June 4, 2024)

Magnetic phase transition in systems with strong electron correlation provides a versatile platform for emergent phenomena. In this work, we extend the newly developed neural network quantum Monte Carlo methods to study magnetic phase transition. We explore the magnetic orders of equispaced hydrogen chains with different interatomic distances, **in both open and periodic systems**. Under external magnetic perturbation, we probe magnetic order from neural network wavefunctions through sampling one- and two-electron reduced density matrices. We find only the $q = \pi$ antiferromagnetic state **exists from strong to weak** correlation regime. High-energy orbitals such as $2s$ and $2p$ are found to be increasingly involved at short interatomic distance, providing guidelines for constructing low-energy effective physics models and for conducting approximated *ab initio* calculations. Our work presents the capability of neural network quantum Monte Carlo methods in studying quantum phases with continuous symmetry.

Understanding quantum phases emerged and phase transitions in interacting many-electron systems have been an interesting yet challenging question in the transdisciplinary field of condensed matter physics and quantum chemistry. Although mean-field theories, mainly density functional theory, has achieved astonishing success, strong correlation systems remain as the main obstacles in the community, which requires more accurate methods. Recent years have witnessed the combination of deep neural network wavefunction *ansatz* with variational Monte Carlo (VMC) in solving complex electron structure [1–3], which is referred to as neural network quantum Monte Carlo (NNQMC). NNQMC methods have been proven promising, providing desirable computational scaling of system size, yet keeping high precision comparable to state-of-the-art methods [1, 3, 4]. Besides the ground state energy, other physical properties have been acquired within NNQMC framework, such as polarization [5], wavefunction localization [3], and radial distribution function in boson Helium bulk [6]. It is now a **legitimate** question whether NNQMC could also probe magnetic phase transition reliably.

Equispaced hydrogen chain is one of the emblematic examples of strongly correlated systems. Simple as it is, the system has long been challenging in calculating ground state energy, and amounts of methods have been tested against this system [7–11]. While previous works have predicted dimerization [12] and Mott transition with respect to interatomic distance [10, 13], magnetic phase transition is another concerned symmetry breaking in hy-

drogen chain, **Where** state-of-the-art theories and numerical calculations only agree on a $q = \pi$ antiferromagnetic phase at large separation limit [10, 13], magnetic order is still unclear for wider range, **at medium to short interatomic distance**. Hydrogen chain is often related to one-band Hubbard and Heisenberg model, which predicts only $q = \pi$ Néel antiferromagnetic (AFM) order for all interacting strength. However, numerical studies with extended terms in model Hamiltonian has it that, ferromagnetic phase [14], various AFM phase, or paramagnetic (PM) phase [15] could be the ground state. For *ab initio* Hamiltonians, only mean-field studies are carried out at wide H atom distance, which suggested ferromagnetic or spiral order [16], and also various different AFM phase [13], for example the forming of magnetic domain.

In this work, using **H-chain as the showground**, we extend NNQMC to study magnetic phase transitions from an accurate *ab initio* perspective. We performed NNQMC calculations for hydrogen chain over a wide range of interatomic distances. To study the phase transition, we implemented sampling of one- and two-electron **RDMs** for both open (OBC) and periodic boundaries (PBC), out of which the one- and two-electron observables can be computed. High-quality energy profile and the dimerization transition are reproduced to demonstrate the accuracy of NNQMC against the state-of-the-art wavefunction calculations. To probe magnetic order, we developed the NNQMC with external magnetic perturbation to compute the magnetic response under finite field and also calculated correlation function at zero-field.

First of all, to demonstrate the accuracy of NNQMC, we compare the energy of equispaced hydrogen chain under OBC with the other state-of-the-art methods (Fig. 1). The equilibrium bond length is around $1.8 a_B$, and the energy minimum is $0.56561(5) E_h$. This result coincides within 1 mE_h with the most accurate results in literature computed using multi-reference configuration interaction with quadruples (MRCI+Q) and auxiliary field quantum Monte Carlo (AFQMC) methods, where results were extrapolated to complete basis set limit. [17]. Aside from the equilibrium bond length, NNQMC also gives good energy results for all the distances calculated, with energy difference less than 1 mE_h . The accurate in energy indicates a reliable ground state wavefunction obtained by NNQMC.

In hydrogen chains at small interatomic distances, due to Peiers instability, it is known that electrons tend to form pairs at neighbouring sites. [18, 19]. Inset of Fig. 1(b) shows a typical dimerization feature at an interatomic distance of $2.4 a_B$. Electron density takes maxima at proton sites, and virtually the same for each atom. Minima of charge density appears at the middle between two sites, and the minima show a periodicity of two site. To quantify the pairing of electron charge density, the dimerization descriptor is defined in eq. (1)

$$P_{\text{dimer}} = \frac{\sum_i |\Gamma_{2i-1,2i}^{(1)} - \Gamma_{2i,2i+1}^{(1)}|}{\sum_i \Gamma_{2i-1,2i}^{(1)} + \Gamma_{2i,2i+1}^{(1)}} \quad (1)$$

$\Gamma_{2i,2i+1}$ and $\Gamma_{2i-1,2i}$ are subdiagonal elements of 1-RDM computed with only $1s$ orbitals. $P_{\text{dimer}} = 1$ stands for a fully dimerized system, while $P_{\text{dimer}} = 0$ means no Peierls pairing. As expected, dimerization grows as the hydrogen stretches [13], which further validates the NNQMC calculation.

Besides the charge density inhomogeneity, antiferromagnetic order also manifests itself in hydrogen chain. At atomic limit, the spins of neighbouring site tend to be opposite, i.e., the neighbouring sites are antiferromagnetically coupled, which results in spontaneous symmetry breaking in spin density. This can be intuitively understood from a way similar to Hubbard chain. Due to the small spatial overlap of two sites, each atom can be considered an atomic magnet to zeroth order approximation. And to the first order, electron with opposite spin can hop across sites and become less localized, hence reducing the kinetic energy, while electrons with the same spin are forbidden by Pauli exclusion and are less favorable in energy. Thus, the antiferromagnetic state is more stable in energy. Actually, for large interatomic distance, hydrogen chain can be approximated by Heisenberg model. For small distance, the approximation to isolated atoms is flawed, and other magnetic order is reported, such as the domain AFM order identified by DFT [13].

To detect the magnetic phase of hydrogen chain, which

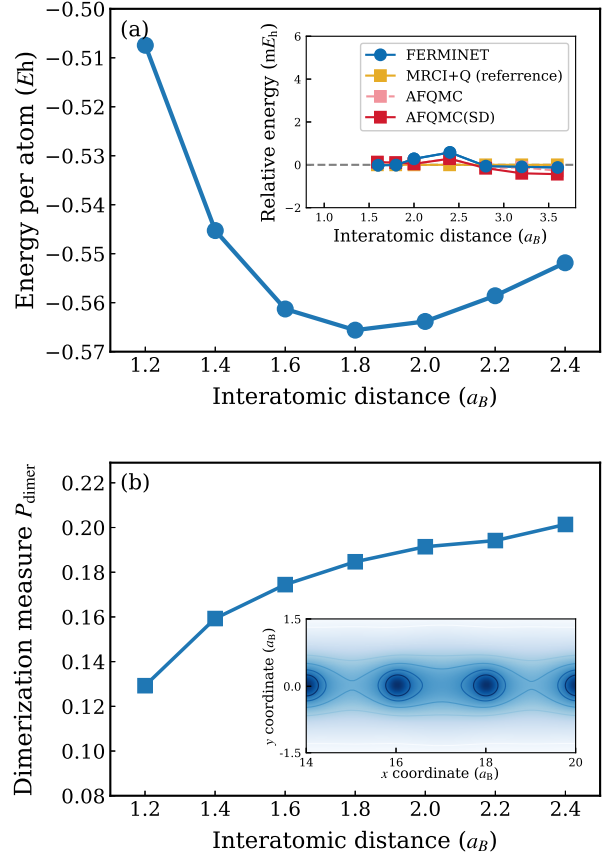


FIG. 1. **Energy and dimerization** (a) Potential energy curve for hydrogen chain with $N=16$ sites at different interatomic distances. Errorbar is negligible, less than $10^{-4} E_h$ for total energy. The inset shows a comparison of the $N=10$ chain with results in ref. [17]. (b) Dimerization measure at different interatomic distances, with $N=16$ and OBC. The inset shows a representative density distribution at $R = 2.4 a_B$, along $x - y$ plane with $z = 0$.

has a $SU(2)$ symmetry and forbids spontaneous symmetry breaking for finite system, we adopt the following two methodologies: (i) apply bulk perturbation to lift degeneracy, and detect directly the order parameter; (ii) calculate the correlation of order parameter between sites, and detect its decay over distance.

Fig. 2 (a) shows a schematic of magnetic field that changes sign for adjacent atoms. The resulting AFM states for various interatomic distance is shown in Fig. 2 (b). The order parameter is chosen to be AFM magnetization $\sum_i |S_i^z|$, and is represented in Fig. 2 (c). By reducing interatomic distance, site magnetic moment reduces. The order parameter approaches zero continu-

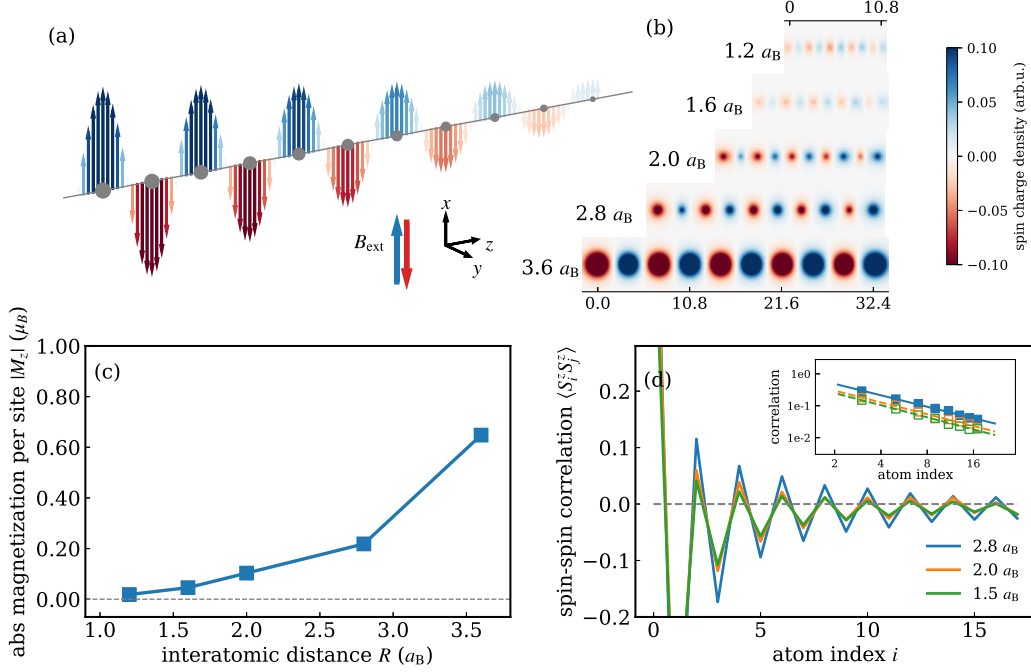


FIG. 2. Antiferromagnetic order in open boundary chains. (a) Schematic plot showing local spin moment pinning with magnetic field applied to each hydrogen site. The blue (red) arrows represent magnetic field pointing to $+x$ ($-x$) directions. (b) spin electron density of $N = 10$ chains, at $R = 1.2, 1.6, 2.0, 2.8, 3.6 a_B$ respectively. The blue (red) shade stands for local spin density with net spin-up (spin-down). (c) total absolute magnetization at different R . (d) decay of correlation function over distance. Inset: critical exponent fitting.

ously, showing no abrupt shift and thus indicates absence of quantum phase transition to PM order.

While the local moment vanishes at zero field, the spin-spin correlator $G_{ss}^z(i-j) \equiv \langle S_i^z S_j^z \rangle$ stays finite. We inspect the decay of this correlator with respect to atom index separation $|i-j|$. Fig. 2 (d) shows NNQMC results of correlation between the left-most and the i -th site, which goes to zeros at long separation as a result of quantum fluctuation. The amplitude of correlation also reduces monotonically, for which we fit to determine the critical exponent, as shown in inset of Fig. 2 (h). The fitting is perform with the form suggested by [20]

$$G_{ss}^z(i) \propto (-)^i i^{-\eta} \ln i, \quad (2)$$

where η is the critical exponent and is predicted to be $\eta = 1$, and the logarithm correction comes from finite size effect. For a $2.8 a_B$ chain, by fitting with difference between even and odd sites, the critical exponent is found to be $1.16(5)$, close to $\eta = 1.11(1)$ result in [13]. The critical exponent η stays unchanged for medium to large distance, indicating absence of phase transitions to other AFM orders in this range. It is worth noting that the correlation amplitude are smaller for even site than odd site. This was also observed in previous DMRG cal-

culations [13], and the even-odd different does not affect the critical exponent. When fitting with even or odd sites separately, the results are $1.2(1)$ for even sites and $1.18(8)$ for odd site.

For cross-validation, we also computed magnetic order in **periodic chains** using NNQMC based on wavefunctions from **DeepSolid** package. Fig. 3 (a) shows the spin-spin correlation function for a 10-atom chain. Again, the correlation function alternates sign for even and odd sites, showing an AFM interaction. The correlation becomes weaker when interatomic distance decreases. Fig. 3 (b) shows the spin structure function, which is directly sampled from real-space electron distribution (see SI section S1), in a similar way to [3]. For all these chains, peaks appears only at $q = \pi$, showing hydrogen chains developed antiferromagnetic order at π .

Lattice models are crucial to insight into low energy behaviours of materials. However, the form of lattice model needs to be carefully chosen. Heisenberg model [20, 21] or Hubbard model [22] are usually used for hydrogen chains, based on which, conformal field theory (CFT) are developed and predicts algebraic decay of correlation over distance. Extension for Hubbard and Heisenberg model usually involves next-to-nearest neighbour hopping, long-

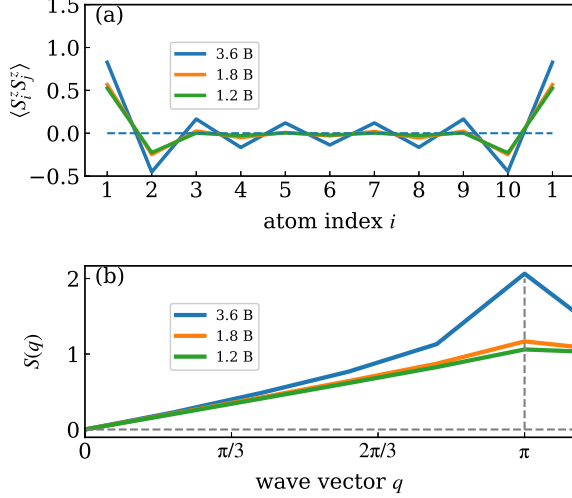


FIG. 3. **Spin structure factor.** Calculation with PBC chain with $N = 10$. (a) correlation for chains distanced by 3.6, 1.8, 1.2 a_B (b) spin structure factor at corresponding distances.

range Coulomb repulsion and multiple orbitals. There are works fitting explicitly low energy Hamiltonian from RDMs with *ab initio* Hamiltonian [23, 24], but alternatively we can exploit RDMs for some intuitions.

One-band models are usually trusted for hydrogen chains, and minimal basis set are commonly used in previous numerical studies on hydrogen chains. NNQMC is basis set free, hence we can now test the adequacy of basis by computing electron occupancy on orbitals. Fig. 4 (a) shows the electron occupation number tested on the 6-31G** basis set. An interesting finding is that while electrons distribute dominantly at 1s-like orbitals, higher-energy orbitals, e.g. 2s and 2p, are also relevant. Their contributions become non-negligible for chains with separation less than 1.0 a_B . Hence, the common one-band Hubbard model scheme for hydrogen chain become insufficient. For even shorter distance, only less than 90% electrons dwell on 1s and 2s. The wavefunction are so diffusive that even 2p orbitals are not enough.

In 4 (b) we show the occupancy of natural orbitals, which are the eigenvalues of the 1-RDM, to further display the strong correlation nature. At small hydrogen distances, the occupation number of natural orbitals are either 0 or 2, indicating no strong Coulomb repulsion for electrons on the same orbitals. When separation increases, the orbitals becomes more isolated so Coulomb interaction surpass interatomic hopping, and natural orbitals tends to be singly occupied. Fig. 4 (c) shows the joint electron density distribution $\langle n_i n_j \rangle$ of site i and j . For both $R = 2.8 a_B$ and $R = 1.5 a_B$ chains, $\langle n_i n_j \rangle$ approaches 1 for next-to-nearest neighbour sites and beyond. Thus, the Coulomb interaction is well screened and affects only the on-site repulsion and nearest-site.

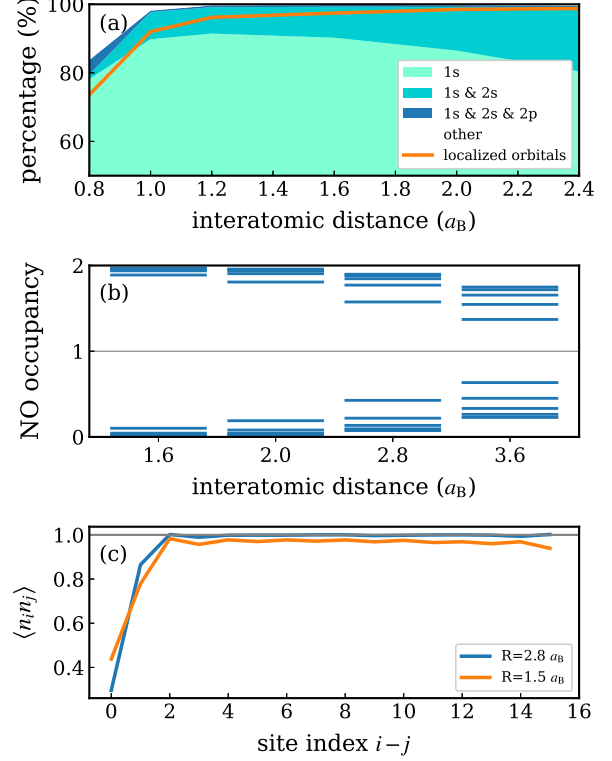


FIG. 4. **Insight into effective model.** (a) Electron occupation of orbitals. The blue shades demonstrates a stackplot of electron number on specific orbitals within 6-31G** basis set, as percentage to total electrons. The orange curve shows electron occupation on the first localized orbitals at each site. (b) Occupation number on natural orbitals. Each vertical line shows occupation number on a certain natural orbital.

In this work, we demonstrate the potential of neural network variational Monte Carlo in capturing quantum phases from **strong correlation regime to weak**. For a finite system with continuous symmetry, we probe the quantum phases by mainly two means, susceptibility to external perturbation, and also the correlators in zero field. These order parameters are computed with implementation of reduced density matrices sampling, in FermiNet package for open boundaries, and DeepSolid for periodic boundary conditions. After demonstrating the capabilities of this method with dimerization, we focused mainly on the magnetic order, especially at medium interatomic distance, which is not fully researched into previously. We calculated the susceptibility with 1-RDM, with alternating magnetic field to **pin** the spin. Also, we calculated **the correlation** from 2-RDM, and fitted critical exponent. We found no abrupt change in spin susceptibility for different bond length, and critical exponent are similar from short to long interatomic distance. These results indicates only AFM phase exists for hydrogen chains with $R > 1.2 a_B$. We also analysed the occu-

pation from RDMs, and found that one-band Hubbard model with on-site repulsion captures the main physics in hydrogen chain, while for short distance, multi-orbital models should be considered instead.

Combining the real space wavefunction with the second quantization representation have several benefits. Firstly, RDM sampling requires much less evaluation of wavefunction than sampling density on spatial grids. In addition, when various observables are needed, only one calculation of RDMs is needed. Beside the spin density and correlation discussed above, RDMs can also be applied for other observables, such as kinetic energy or spectrum density [25]. Moreover, the same methodology are compatible with transition RDMs, which can be related to photo emission. When compared with RDMs from other second quantization methods, here the basis set are chosen for RDMs only, and does not affect the optimization of the wavefunction. Thus, the basis set incompleteness can be measured by RDMs, unlike many second quantization methods, where a basis set is chosen *a priori* by testing energy convergence. Therefore, we can envision a broad range of applications of our methods to study quantum phase transitions and chemical transitions.

This work was supported by the National Key R&D Program of China under Grant No. 2021YFA1400500, the Strategic Priority Research Program of the Chinese Academy of Sciences under Grant No. XDB33000000, and National Science Foundation of China under Grant No. 12334003. We are grateful for computational resources provided by the High Performance Computing Platform of Peking University.

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S1 Wavefunction sampling algorithms

RDM sampling algorithm and extension to periodic systems

Here we define 1-RDM and 2-RDM with notation that

$$\Gamma_{ij}^{(1)} = \langle \Psi | c_i^\dagger c_j | \Psi \rangle \quad (3)$$

$$\Gamma_{ijkl}^{(2)} = \langle \Psi | c_i^\dagger c_j^\dagger c_l c_k | \Psi \rangle \quad (4)$$

where $c_i^\dagger(c_j)$ is creation (annihilation) operator of the i -th (j -th) orbital. For a many-body wavefunction $|\Psi\rangle$, operator $c_i^\dagger c_j$ moves an electron from j -th orbital to i -th orbital.

Here, following method in ref. [14], RDMs are sampled with

$$\Gamma_{ij}^{(1)} = \sum_a \frac{\langle \frac{\Psi(R'_a)}{\Psi^*(R)} \frac{\varphi_i^*(r_a) \varphi_j(r'_a)}{f(r'_a)} \rangle_{|\Psi|^2, f}}{N_i N_j} \quad (5)$$

$$\Gamma_{ijkl}^{(2)} = \sum_{a \neq b} \frac{\langle \frac{\Psi(R'_{ab})}{\Psi^*(R)} \frac{\varphi_i^*(r_a) \varphi_j^*(r_b) \varphi_k(r'_a) \varphi_l(r'_b)}{f(r'_a) f(r'_b)} \rangle_{|\Psi|^2, f}}{N_i N_j N_k N_l} \quad (6)$$

where $R'_a = R : r_a \rightarrow r'_a$, $R'_{ab} = R : r_a \rightarrow r'_a, r_b \rightarrow r'_b$. The subscript $|\Psi|^2, f$ denotes importance sampling for r'_a and r'_b over

$$f(r'_a) = \sum_i |\varphi_i(r'_a)|^2, \quad (7)$$

and $|\Psi(R)|^2$ for $r_1 \cdot r_N$. N_i is normalization factor for φ_i with

$$N_i = \langle |\varphi_i(r'_a)|^2 / f(r'_a) \rangle_f^{1/2}. \quad (8)$$

These relations can be derived with the relation $c_i^\dagger = \int dr \hat{\varphi}^\dagger(r) \varphi_i(r)$. we have $c_i^\dagger |r_1, r_2, \dots, r_N\rangle = \sum_a \varphi_i(r_a) |r_1, r_2, \dots, r_a, \dots, r_N\rangle$. Further we have

$$\langle \Psi | c_i^\dagger c_j^\dagger | \Psi \rangle = \sum_a \int d\{R\} dr' \varphi_i(r_a) \varphi_j^*(r'_a) \times \Psi^*(R) \Psi(R : r_a \rightarrow r'_a) \quad (9)$$

where $dR = dr_1 dr_2 \dots dr_N$, and $R : r_a \rightarrow r'_a = \{r_1, r_2, \dots, r'_a, \dots, r_N\}$. Thus, the sampling of 1-RDM is an $(M+1)$ -dimensional integral. Similarly, 2-RDMs can be sampled with an $(M+2)$ -dimensional integral.

$$\langle c_i^\dagger c_j^\dagger c_l c_k \rangle = \sum_{a \neq b} \int d\{R\} dr'_a dr'_b \varphi_i(r_a) \varphi_j(r_b) \varphi_k(r'_a) \varphi_l(r'_b) \times \Psi^*(R) \Psi(R : r_a \rightarrow r'_a, r_b \rightarrow r'_b) \quad (10)$$

To speedup convergence, importance sampling technique is applied. Metropolis process are performed with importance $|\Psi|^2$ for R , and r'_a, r'_b are sampled with $f(r) = \sum_i |\varphi_i(r)|^2$. and the integral is modified as eq. 5 and eq. 6. Further, normalization for φ_i are imposed by evaluating of integral $|\varphi_i|^2$, with importance weight $f(r)$, which lead to eq. 8.

For non-orthogonal basis, RDMs are contravariant under transformation $\varphi_i \rightarrow C_I^i \varphi_i$, 10 transforms as

$$\langle \Psi | c_i^\dagger c_j^\dagger | \Psi \rangle = \sum_a \int d\{R\} dr' \tilde{C}_i^I \varphi_i(r_a) \tilde{C}_j^J \varphi_j^*(r'_a) \times \Psi^*(R) \Psi(R : r_a \rightarrow r'_a) \quad (11)$$

with $\tilde{\mathbf{C}} \mathbf{S} \mathbf{C} = \mathbf{1}$, $\mathbf{S}_{ij} = \langle \varphi_i | \varphi_j \rangle$.

For periodic boundary condition, the integral eq. 5 should be evaluated within a single primitive cell. φ_i is localized orbital and is not periodic, and for an electron at r'_a outside the primitive cell, we should evaluate Ψ by shifting the a -th electron with lattice constant. By utilizing $\varphi_i(r'_a + a) = \varphi_{i-1}(r'_a)$, we arrive at the sampling for periodic system

$$\langle \Psi | c_i^\dagger c_j^\dagger | \Psi \rangle = \sum_a \int_{\text{cell}} d\{R\} dr' \sum_{T=-\infty}^{\infty} \varphi_i(r_a + aT) \sum_{j=-\infty}^{\infty} \varphi_j^*(r'_a + aT) \times \Psi^*(R) \Psi(R : r_a \rightarrow r'_a), \quad (12)$$

where a is the lattice vector, and T is the index of periodic images. The evaluation of orbital i should in theory contain all the periodic images, and we take a real space cutoff. It should be noted that, only 2 adjacent cells are not enough. For orbitals centered at the boundary of primitive cell, at least 27 neighbouring images should be considered to will lead to a wrongful distribution of r'_a and hence wrongful RDMs.

Error analysis of RDM sampling

We check the stochastic error of RDMs with block analysis technique, performed with pyblock package. Within a 20000-iteration simulation, the maximum standard error is estimated as Table . And also, RDMs are Hermitian in theory, and obey fermion permutation antisymmetry, which are not forced in RDM sampling process. We estimated error as Table

TABLE I. maximum std error of RDMs, N=18

system	$\Gamma_{aa}^{(1)}$	$\Gamma_{bb}^{(1)}$	$\Gamma_{aaaa}^{(2)}$	$\Gamma_{abab}^{(2)}$	$\Gamma_{bbbb}^{(2)}$
R=1.0 B	0.000318	0.001519	0.000907	0.000915	0.000879
R=3.6 B	0.000346	0.000347	0.001134	0.001136	0.001144

TABLE II. maximum non-hermitian error of RDMs, N=18

system	$\Gamma_{aa}^{(1)}$	$\Gamma_{bb}^{(1)}$	$\Gamma_{aaaa}^{(2)}$	$\Gamma_{bbbb}^{(2)}$
R=1.0 B	0.001283	0.001519	0.004432	0.004205
R=3.6 B	0.001695	0.001475	0.003403	0.003699

Spin structure factor

The spin structure factor is defined as

$$S(q) = \frac{1}{N} \langle \rho_z(q) \rho_z(-q) \rangle - \langle \rho_z(q) \rangle \langle \rho_z(-q) \rangle, \quad (13)$$

where $\rho_z(q)$ is Fourier transformation of spin density $\rho_\uparrow(r) - \rho_\downarrow(r)$. The evaluation is performed with Monte Carlo method, for every Monte Carlo step, $\rho_z^\uparrow(q)$ is sampled with $\sum_{i \in \text{spin}=\uparrow} e^{iqr_i}$, and thus,

$$S(q) = \langle (\rho_\uparrow(q) - \rho_\downarrow(q))(\rho_\uparrow(-q) - \rho_\downarrow(-q)) \rangle \quad (14)$$

$$- \langle \rho_\uparrow(q) - \rho_\downarrow(q) \rangle \langle \rho_\uparrow(-q) - \rho_\downarrow(-q) \rangle \quad (15)$$

$$= \langle (\rho_\uparrow(q) - \rho_\downarrow(q))(\rho_\uparrow^*(q) - \rho_\downarrow^*(q)) \rangle \quad (16)$$

$$- \langle \rho_\uparrow(q) - \rho_\downarrow(q) \rangle \langle \rho_\uparrow^*(q) - \rho_\downarrow^*(q) \rangle \quad (17)$$

Choice of localized orbitals

In the main text we use meta-Löwdin methods for dividing sites, and there're other methods for orbital local-

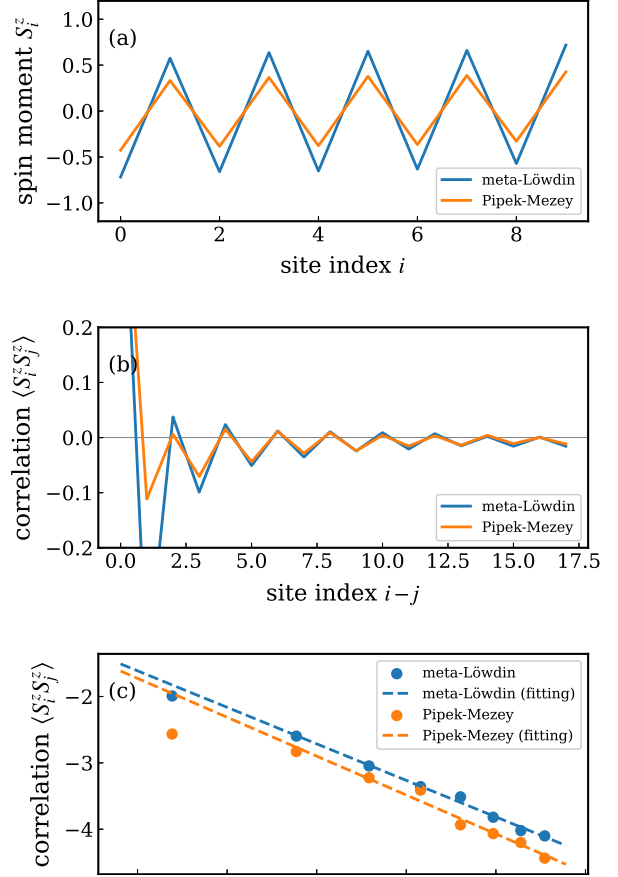


FIG. 5. **Antiferromagnetic order result for different localized orbitals.** (a) site spin magnetization moment for OBC $N = 10$, $R = 3.6 a_B$ chain, with different localization methods. (b) correlation function for OBC $N = 18$, $R = 1.5 a_B$ chain. (c) critical exponent for OBC $N = 18$, $R = 1.5 a_B$ chain.

ization. The result are shown in 5. In 5 (a), the magnetic moment in each site are slightly different for two localization methods, but still these two methods capture magnetization well. And also, in main text total absolute spin density are used as order parameter, and integration are carried out for localized orbitals and is not affected by the localization method. In 5 (b) and (c), we show the correlation function and fitting of critical exponent. Although the value of correlation function changes, the critical exponent is not much affect.

S2 Further computational details

inclination in AFM order

In the calculation with bulk perturbation, we found that the calculated state shows a deviation among sites in

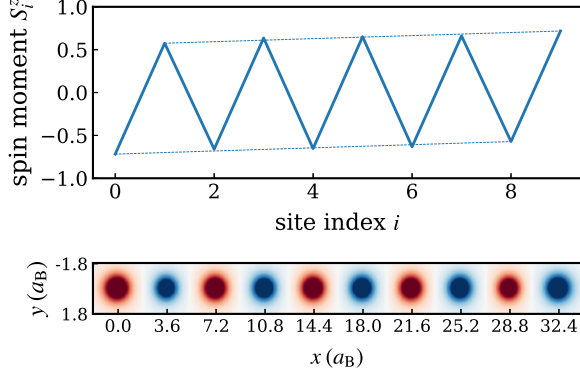


FIG. 6. **Antiferromagnetic order in finite chain.** (a) spin density distribution of $N = 10$ chain with $R = 3.6 a_B$ (b) site magnetic moment in (a).

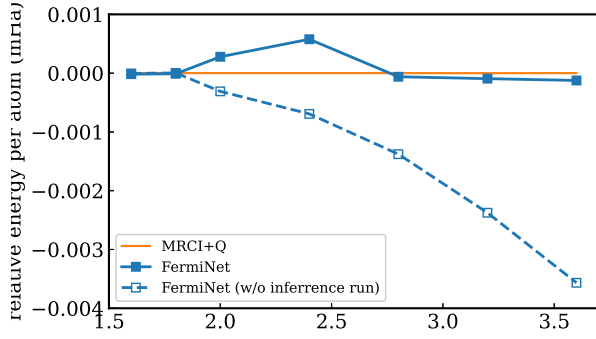


FIG. 7. **Overshooting of FermiNet.**

local magnetic moment. Site magnetic moment is shown in Fig. 6, and shows a systematic inclination from left to right, which we assumed a result of the gapless spin wave mode. Here, the inclination in site magnetic moment doesn't affect the continuous change of absolute magnetization, and thus the absence of AFM-PM phase transition is not affected. Also, spin-wave mode are expected only for AFM state, and may imply an AFM for all the bond length presented.

overshooting of FermiNet

Overshooting are found when not enough steps are performed between optimization, as the MCMC process are not balancedly distributed and cause bias, as shown in Fig. 7. This bias can be avoided by inference run, i.e. averaging observables after the wavefunction optimization completed. All the energies and RDMs are sampled after training process, to avoid potential overshooting bias.

external field effects

We tested the strength of external field by applying to single atom. For an external Gaussian potential $B_{ext}(r) = e^{-(r-R_0)^2/2\sigma^2}$, where R_0 is the nuclear, and with $\sigma = 0.1 a_B$, field strength $0.5E_h$. The total energy is $-0.501845(8) E_h$. Also, we inverse the field direction, and keep the wavefunction unchanged, the energy is $-0.498190(15) e_h$. Thus the external does not significantly change the ground state wavefunction.